How Deep Does a Next-Concept Model Need to Run? Measuring the Depth Ceiling of Coarse-to-Fine Text Generation

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Abstract

Next-concept models write a sequence in a fixed number of coarse-to-fine passes: K passes for every input, whether it needs them or not. We ask how many passes a task actually requires, and we answer it before building a rule. We define the *oracle depth ceiling*, the share of the ladder that a rule with perfect foresight would skip at equal quality, and measure it across tasks: 37.7 percent on XSum summarisation, repeating at 39.3 and 37.6 percent on two unseen cohorts, 6.7 percent on 6x6 Sudoku, and zero on 9x9 Sudoku, where no board is correct before the last level. The number depends on the meter: the same passages allow 6.2 to 7.4 percent once the meter counts content words against the model's own full-depth output, because the last pass leaves the length unchanged and replaces one content word in seven. A training-free rule that stops when the output's named entities stop changing captures a third of the summarisation ceiling, saving one pass in eight at 0.20 to 0.62 ROUGE-L points against a declared allowance of 1.0, and ahead of a constant-depth cutoff at matched cost. A calibrated-confidence gate is dominated: it spends more passes and loses more quality and more entity recall. On a third cohort the content budget fails: entity recall drops 1.00 point against a budget of 0.5, and ROUGE-L does not see it. Early exit is cheap in the metric it optimises and not free in content.

1 Introduction

A next-concept model writes a sequence in K passes. The first pass emits one code, the second two, and so on up a ladder until the last pass emits every position. Each pass predicts all of its codes in parallel, so the model is cheap compared with one-token-at-a-time decoding, and the ladder is fixed at design time [1]. Every input climbs the whole ladder, whether it is a one-line summary or a page of markup.

That raises a question the model's design cannot answer. How much of the ladder is needed?

Two conditions must hold before an exit rule can help. The output at an early level must be good enough to keep, and the moment when it becomes good enough must be visible from the output alone, without the reference. When either condition fails, the saving is zero and no amount of tuning changes that. This paper measures both conditions first, then builds a rule, and reports what the rule costs.

1.1 The ceiling comes before the rule

We define the *oracle depth ceiling*: the share of the ladder that a rule with perfect foresight would skip while keeping output quality. On XSum summarisation the ceiling is 37.7 percent, and it repeats at 39.3 and 37.6 percent on two further disjoint cohorts of 500 passages. On 9x9 Sudoku the ceiling is zero: no board is correct before the last level, so every board must climb the whole ladder. On 6x6 grids it is 6.7 percent.

The model family is the same in all three settings, and the ladder design is the same. The ceiling moves from zero to more than half the ladder because of the task and its metric, not because of the model.

This is the part of the problem that a gate comparison cannot supply. A rule can look good without being close to the best possible, and the literature rarely reports how far away it is. The ceiling gives the denominator.

1.2 A rule with nothing to calibrate

We then ask whether a simple observable can reach part of the ceiling. The observable is the set of names in the output. Exit at the first level, at or after a minimum, whose entity set is a subset of the previous level's, for two consecutive levels. If names are still arriving, keep going; when they stop, the content is complete and further passes only re-word it.

The rule needs no training, no threshold, and no reference. It reads text the model has already produced.

On two cohorts it saves one pass in eight, 12.4 and 13.0 percent of the ladder, at a ROUGE-L cost of 0.62 and 0.29 points against a declared allowance of 1.0. That is about a third of the ceiling. At matched cost it beats stopping every input at a constant depth. Neither gate we tested survives: a calibrated-risk gate, which loses more quality than the constant cutoff it was meant to improve, and a source-coverage gate cannot fire at all, because a 48-token summary holds 3.3 percent of the names in a 512-token article against a 50 percent bar.

1.3 What the saving costs

A third cohort, frozen before it was decoded, changes how the result should be stated. The cost saving and the ROUGE-L cost reproduce: 12.2 percent saved, 0.20 points lost. Entity recall does not. It falls 1.00 point against a declared budget of 0.5, with an interval that excludes zero.

So the honest claim is narrow and two-sided. Early exit at a graded metric is close to free in the metric it optimises, and it is not free in content: the summary keeps its length and its score while losing names, at between 0.3 and 1.0 recall points per pass saved. ROUGE-L does not see the loss, which is why the content meter matters and why the source-coverage gate was the wrong instrument.

The NCM paper documents the small version of the same failure: its tokenizer can replace one named entity or numeral with another while the sentence stays fluent [1]. Here the same symptom appears at a larger scale, with the codebook held fixed.

1.4 Contributions

1. **A measured depth ceiling.** An oracle definition, reported on three XSum cohorts, two grid tasks, and one document task. The ceiling is a property of the task, the metric, and the tolerance, and it moves by a factor of six across settings (Section 4.2).

2. **A training-free exit rule that reaches a third of it.** Entity saturation, with no head, no threshold, and no reference, compared against two gates and a constant-depth cutoff at matched cost on three cohorts (Sections 4.3 and 4.4).

3. **A content cost that ROUGE does not show.** A paired entity-recall audit that separates a saving that reproduces from a budget that does not, with the failure rate reported beside the saving (Section 4.4).

4. **The meter slope and the pass that causes it.** The last pass leaves the length unchanged and replaces one content word in seven, so the ceiling falls from 37.7 to 6.4 percent on the same passages when the meter counts content words rather than reference agreement (Section 4.5).

5. **A diagnostic for new tasks.** Measure the per-level exactness or content curve before building a gate. A task whose output never becomes complete below the top level has a ceiling of zero and no exit rule can help (Section 4.6).

1.5 Scope

One model family, one text corpus in three slices, greedy decoding, and no training. The text result is a claim about the released XSum checkpoint, not about scale. The grid and document results establish the ceiling boundary and the tokenizer's ability to support a saving; they are not claims about generation quality on those tasks.

The rule saves 12 to 13 percent, and 12 to 13 percent of passages come out materially worse. A reader who needs every passage to hold its quality should not use it. The contribution is the measurement and the honest price, not a free lunch.

2 Related work

2.1 Next-concept generation

Next-concept models (NCM) generate a target through a learned hierarchy of discrete codes, coarse to fine, one level at a time [1]. Each level predicts all of its codes in one forward pass under block-diagonal attention, so a sequence of length L is produced in K passes instead of L. The family descends from two lines: discrete latent tokenizers, VQ-VAE and its residual variants [2, 3, 4], and next-scale prediction, which showed for images that coarse-to-fine parallel decoding is both faster and competitive in quality [5]. NCM carries both to text, where the coarse levels have no natural counterpart to image downsampling and where a single wrong token can change a fact, break code, or invalidate a board [1].

The step count of such a model is fixed by design: K levels, every input. That is the property this paper questions.

2.2 Adaptive computation and early exit

Static depth means the same computation for every input. Dynamic depth varies it per input, and has a long history. Adaptive Computation Time learns a halting unit that adds computation until a cumulative probability crosses a threshold [6]. Early-exit networks attach classifiers to intermediate layers and stop when a confidence test passes, from BranchyNet onward [7], and the same idea is used to skip layers in large transformers [8, 9]. Cascade systems ask the same question from the systems side: run the cheap model, and escalate only when it fails a test [10].

Two lessons from that literature shape this paper. First, the decision rule is usually a threshold on a confidence signal, and the threshold is a tuned hyperparameter. Second, the saving is reported but the ceiling is not: the literature rarely asks how much saving the task allows in principle, so a rule can be called successful without anyone knowing how far it is from the best possible.

Speculative decoding takes a different route to the same goal — spend less compute per token — by drafting and verifying instead of stopping early [11]. It is orthogonal to depth selection and could be combined with it.

2.3 Anytime and progressive generation

Iterative generators already produce partial outputs. Diffusion and flow models over text refine a full sequence at every step [12, 13, 14], and a body of work reduces the number of steps by distillation or learned schedules [15]. Those methods shorten the trajectory for all inputs. NCM's ladder is different: its levels are nested prefixes of the target, so an early level is a valid but shorter output, not a partially denoised one. That nesting is what makes an exit rule possible at all, and it is also what makes the ceiling measurable — there is a well-defined "output at level k" to score.

2.4 What the metrics do not see

Summaries are scored with ROUGE [16], which counts n-gram overlap and is known to track surface form more reliably than meaning [17]. Factuality work replaced it with entailment and question-answering checks [18, 19], and evaluation suites for abstractive summarization measure whether the entities and numbers survive [20, 21].

This paper uses that split deliberately. It reports ROUGE-L as the quality endpoint and entity recall as the content endpoint, and it reports when the two disagree. They do, on one of the three cohorts.

The NCM paper itself documents the small version of the same failure. Its tokenizer appendix lists reconstruction errors that substitute named entities and numerals in otherwise fluent text, for example a city name replaced by another city name [1]. Section 4.4 reports the same behaviour at a larger scale, caused by depth rather than by the codebook.

2.5 What this paper adds

Three things are missing from the work above, and each is a contribution here.

1. **A measured ceiling.** The oracle depth ceiling answers how much of the ladder any rule could skip, before any rule is built. It is a property of a task, a metric, and a tolerance, and it moves from zero to 54.6 percent between summaries and grids.

2. **A rule that needs no training and no threshold.** Entity saturation compares the output to itself, so there is nothing to calibrate. It reaches a third of the ceiling, and it is reported against two gates and a constant-depth cutoff at matched cost.

3. **A diagnostic that runs before the model.** Measure the per-level exactness or content curve. If the output never becomes complete below the top level, the ceiling is zero and no exit rule can help. This decides in an afternoon whether a task is worth the effort.

3 Method

3.1 Setting

A next-concept model generates one target in K passes over a fixed ladder of positions $\ell_0 < \ell_1 < \cdots < \ell_{K-1}$. Pass k reads the reconstruction produced by pass $k-1$ and emits ℓ_k codes, so the output after pass k is the first ℓ_k tokens of the target. Nothing in the model changes between passes. The ladder is a property of the tokenizer, fixed before training and never tuned here.

The cost of one output is its number of passes, NFE. A decode that stops after pass k has used $k+1$ passes. The full ladder costs K.

Throughout, $k_{\min}$ is the earliest level a rule may stop at. On XSum, $K = 9$, $k_{\min} = 3$, and the ladder is 4, 6, 10, 14, 20, 26, 32, 40, 48 tokens. The decode is greedy, so a given input and checkpoint always produce the same trajectory.

3.2 The ceiling

Let $r_{i,k}$ be the ROUGE-L of input i's prefix after pass k, and $r_{i,*}$ its value at the last pass. A materiality tolerance $\tau = 0.005$ marks a difference smaller than half a ROUGE point as immaterial; this is the same bar used to count a passage as materially worse.

The oracle for input i stops at the smallest $k \geq k_{\min}$ with

$$r_{i,k} \;\geq\; r_{i,*} - \tau, \tag{1}$$

and at the last pass when no such k exists. The ceiling is

$$C \;=\; 1 - \mathrm{mean}_i\big(\mathrm{NFE}_{\mathrm{oracle}}(i)\big)/K, \tag{2}$$

which is the share of the ladder that a rule with perfect foresight would skip.

The ceiling is an upper bound for any rule that scores by ROUGE-L, because any such rule must choose without seeing the final score. It is not a bound for other meters. A rule can be worse than the oracle on ROUGE-L and better on content, and the reverse. Write the ceiling as a property of a triple: task, meter, tolerance. It is not a property of the task alone.

3.3 The content meter

Entity recall asks whether the summary still names the things the reference names.

For a text t, $E(t)$ is the set of names in t under a capitalization census rule:

- a run of capitalized tokens is a candidate span, and spans joined by a connector are one span;
- leading articles are removed;
- a capitalized token that is not sentence-initial counts as a name only if it also appears capitalized mid-sentence somewhere in the corpus (`mid_caps`), which separates ordinary words that open a sentence from proper names;
- a sentence-initial token that is not in `mid_caps` is not a name;
- numbers and month names are names.

For input i with generated prefix y and reference y^*,

$$\mathrm{recall}_i \;=\; \frac{|E(y) \cap E(y^*)|}{|E(y^*)|}, \tag{3}$$

and the reported recall is the mean over inputs with a non-empty $E(y^*)$. The paired difference against the fixed-depth arm uses the same inputs, and the interval is a 10,000-sample bootstrap over passages.

`mid_caps` is compiled once per corpus from the sources, the references and the generated outputs of the same cohort. It decides only whether a capitalized token is a name. It is not tuned on any quality outcome, and no reported number is selected by it. It is nevertheless a corpus-level object, and a deployment needs to build one.

3.4 The rule

Entity saturation watches the output, not the source, and needs no reference.

For a trajectory, let $E_k = E(\text{prefix}_k)$. With quiet parameter q, the rule stops at the smallest level $k \geq k_{\min}$ such that the q most recent level pairs all satisfy

$$E_j \subseteq E_{j-1} \qquad \text{for } j \in \{k-q+1, \ldots, k\}, \tag{4}$$

and otherwise runs to the last pass. The frozen setting is $q = 2$.

Subset, and not a count. The number of names can stay flat while the set changes: one name leaves as another arrives. A count-based rule would read that as quiet and stop at the wrong moment. The subset test asks the question that matters, which is whether any name has appeared that was not there before.

Two consecutive quiet levels, and not one. A single non-arrival can be a span that scores one pass later after a boundary decision changes. Two in a row is the cheapest guard against that, and $q = 1$ is reported beside the frozen rule as the cost of not having the guard.

3.5 Cost of the rule

The rule reads text that the decode has already produced. There is no extra forward pass, no head, and no model.

Measured by `results/rule_cost.py` over 500 passages and all nine levels: 2.2 ms per passage on one CPU core, for extraction, set construction and the subset test together. Generation costs 3.2 s per passage on the CPU that ran the replay. Even allowing a factor of ten in either direction, the rule stays well under one percent of the compute it decides about.

The same script reproduces the arm's mean NFE within two passages of the recorded value. The difference comes from the census vocabulary: the replay compiles it from passages, references and generated text, and the script from generated text alone. The cost figure does not depend on that choice.

The only object that is not per-input is `mid_caps`. It is linear in the corpus, built once, and shared by every input.

3.6 What is trained

Nothing. The checkpoint is the released XSum generator, the tokenizer is frozen, and the decode settings are fixed. The rule has one parameter, q, chosen on the first cohort and then confirmed unchanged on two later cohorts.

This matters for how the result should be read. The saving is available without a training run, without a gate head, and without labels for the task. It also means the rule cannot adapt to a distribution it has not seen: the census vocabulary and q are the only places where a new corpus could require work.

3.7 Why not a gate

A gate maps a continuous signal to a decision through a threshold. Two natural signals were tested and reported in Section 4.3.

A calibrated-risk gate needs the risk estimate to stay meaningful across inputs and levels. It does not: at matched cost the gate loses more quality than a constant-depth cutoff, and on Sudoku it stops early and solves 1 board of 200.

Table 1: The oracle depth ceiling by task. Oracle passes is the mean number of passes a rule with perfect foresight uses; fixed passes is the full ladder. Saved is the share of the ladder the oracle skips. † The document row is the tokenizer's own oracle on 200 held-out documents; no generator was trained for that task, so it measures the codes, not a policy.

task	n	ladder	oracle passes	fixed passes	saved
XSum, cohort A	500	9	5.61	9.00	37.7%
XSum, cohort B	500	9	5.46	9.00	39.3%
XSum, cohort C	500	9	5.61	9.00	37.6%
HTML documents†	200	10	4.54	10.00	54.6%
Sudoku 6x6	200	10	9.33	10.00	6.7%
Sudoku 9x9 hard	200	12	12.00	12.00	0.0%

A source-coverage gate needs the partial output to hold a fixed share of the source's names. At 48 tokens a summary holds 3.3 percent of them, and the best passage at any level reaches 25 percent against a 50 percent bar. The gate cannot fire before any tuning starts.

The saturation rule avoids both failure modes because it compares the output to itself. There is no threshold to calibrate and nothing to normalise across lengths.

4 Results

4.1 Setup

Model: the released XSum generator, about 99M task-specific parameters, 12 layers, width 768, prefix 512 tokens. The VQ tokenizer is frozen. Decode is greedy: temperature 0, frequency penalty 2.0, adjacency penalties 5, 4, 0.5. The ladder holds 4, 6, 10, 14, 20, 26, 32, 40, 48 tokens at levels 0 to 8. $K = 9$, and the earliest permitted exit is $k_{\min} = 3$. No weights are trained in this section.

Three disjoint cohorts of 500 XSum test passages. Cohort A is passages 0 to 499, used to choose the rule. Cohort B is passages 500 to 999; it was never decoded before the rule in Section 4.4 was written down. Cohort C is passages 1000 to 1499, built after B and frozen in `results/saturation_prereg_c_2026-10-27.md` before it was decoded, as a replication. The probe records the sha256 of both pickles, and the replay checks it, so the scripts that produce every number below read the same 500 passages.

Quality is ROUGE-L against the reference summary. Content is the recall of named entities found in the reference, under a capitalization census rule. Two bars were declared before the runs: a mean allowance of 0.01 ROUGE-L, and a per-example materiality bar of 0.5 ROUGE-L points. A loss below the per-example bar is not a false exit.

4.2 The ceiling belongs to the task

A perfect exit rule knows, for each input, the first level whose output is as good as the last. Table 1 gives that oracle for six settings.

The ceiling spans the whole range, from zero on 9x9 grids to 54.6 percent on documents. On summaries, an ideal rule would save more than a third of the passes. On 6x6 grids it saves one pass in fifteen. On 9x9 grids it saves nothing: no board is exact at any level below the top, so every board must run the full ladder.

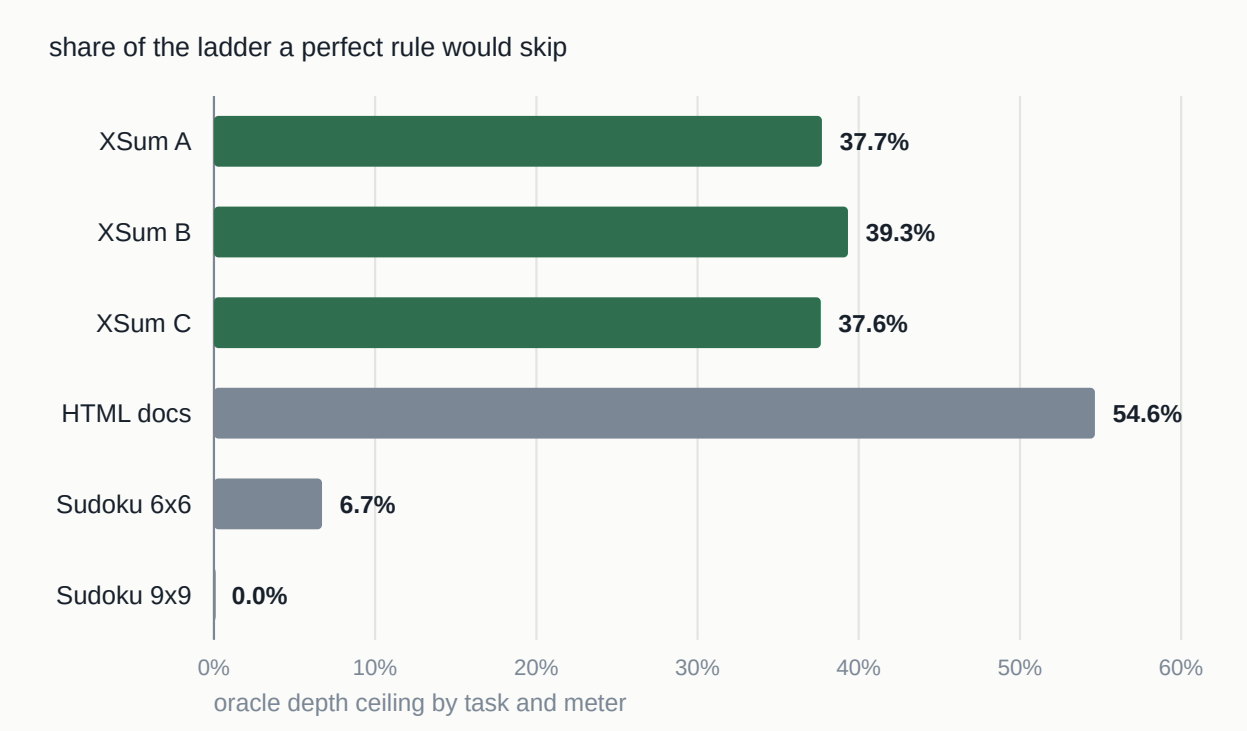


Figure 1: The oracle depth ceiling by task. For each setting, the passes a rule with perfect foresight uses against the fixed ladder. The ceiling is largest on documents, more than a third on summaries, one pass in fifteen on 6x6 grids, and zero on 9x9 grids.

Table 2: The calibrated-risk gate on cohort B, against the fixed ladder. Intervals are 95 percent passage bootstraps. A passage is materially worse when it loses more than 0.5 ROUGE-L points.

arm	passes	ΔROUGE-L (pts)	Δentity recall (pts)	materially worse
fixed	9.00	control	control	0%
α 0.70	7.05	-1.08 $[-1.42, -0.75]$	-5.42 $[-7.24, -3.69]$	27.8%
α 0.85	4.29	-2.54 $[-3.13, -1.98]$	-11.64 $[-13.95, -9.43]$	51.4%

The model is the same kind of model, and the ladder is the same design. The task sets how early the content settles. The meter sets how much of what is left may be skipped, and Section 4.5 measures that second half. Figure 1 plots the same settings.

4.3 Two gates that miss the ceiling

Confidence. Exit at the first level at or after $k_{\min}$ whose token-margin risk falls below α. On cohort B this gate is worse than doing nothing.

It also loses to a constant-depth policy. On cohort A the gate reaches 7.02 passes at 0.2375 ROUGE-L, while stopping every passage at level 6 gives 0.2403 at 7.00 passes. A per-input rule that is beaten by a fixed cutoff does not earn the machinery.

On Sudoku the same gate fails without ambiguity. At α 0.10 it stops after 5.07 passes and solves 1 of 200 boards. Fixed depth solves 169 of 200.

Source coverage. Exit when the partial summary already contains a fraction R of the entity

Table 3: Entity saturation on three cohorts. Cohort A chose the rule; B was pre-registered; C is a replication frozen before it was decoded. Intervals are 95 percent passage bootstraps. Costs are in ROUGE-L and entity-recall points; worse and better are the share of passages losing or gaining more than 0.5 ROUGE points against fixed depth.

cohort	passes	ΔROUGE-L	Δrecall	worse	better
A, explored	7.89	−0.62	−0.24	16.0%	7.4%
B, prereg.	7.83	−0.29 [−0.53, −0.06]	−0.31 [−1.22, +0.56]	13.0%	8.8%
C, replication	7.90	−0.20 [−0.44, +0.03]	−1.00 [−1.96, −0.17]	11.6%	9.2%

spans in the source prefix. This gate never fires. A full-depth summary covers 3.3% of the source entities, and the best passage at any level reaches 25%, against a minimum bar of 50%. The gate is unreachable before any tuning starts, because a 48-token output cannot hold the names of a 512-token article.

4.4 One gate that works on cost, and what it costs in content

The rule watches the output, not the source. Exit at the first level at or after $k_{\min}$ whose entity set is a subset of the previous level's entity set, for two consecutive levels. Otherwise run to the ladder end.

The rule needs no reference and no threshold on coverage. It asks one question: are new names still arriving?

Three bars were registered before cohort B ran: mean passes below 9.00, a ROUGE-L loss no worse than 1.0 point, and an entity-recall loss no worse than 0.5 point. Cohorts B and C both pass the first two. Neither passes the third. On B the recall loss is 0.31 points and its interval contains zero; on C it is 1.00 points, twice the allowance, with an interval that excludes zero. Pooled over the two cohorts the loss is 0.66 points, so the budget is missed in aggregate as well. In the fraction units of the registered bars, B returns -0.0029 $[-0.0053, -0.0006]$ on ROUGE-L against a bar of -0.0100, C returns -0.0100 $[-0.0196, -0.0017]$ on entity recall against a bar of -0.005, and the pooled B and C recall value is -0.0066.

The saving is not in question. The rule skips one pass in eight on both cohorts, and the ROUGE-L loss is between a fifth and a third of the declared allowance. It beats a constant cutoff at matched cost on both cohorts: at 7.83 passes, interpolated fixed depth gives 0.2465 against the rule's 0.2502, and at 7.90 passes on cohort C the same comparison gives 0.2421 against 0.2478, a gap of 0.57 points. On cohort C it also beats both alternatives at their own budgets: α 0.70 reaches 6.94 passes at a cost of 1.34 ROUGE points and 6.60 recall points, and `sat_1` reaches 6.57 passes at a cost of 0.95 and 3.63.

What the third cohort establishes is the shape of the price. The summary keeps its length and its ROUGE score while losing names, at between 0.3 and 1.0 recall points per pass saved. ROUGE-L does not see this, and a reader who trusts it alone would call the rule free. The same effect kills the source-coverage gate in Section 4.3, measured there from the source side.

Figure 2 places every arm on both planes at once. The quality panel shows the rule sitting above the constant-depth curve; the content panel shows the same arms falling further, because recall drops faster than ROUGE.

Two limits belong beside the number. The choice of two consecutive quiet levels was made on cohort A, so B and C confirm a chosen rule rather than discover one. And 58% of passages still need all nine passes; the saving comes from the passages whose names arrive early.

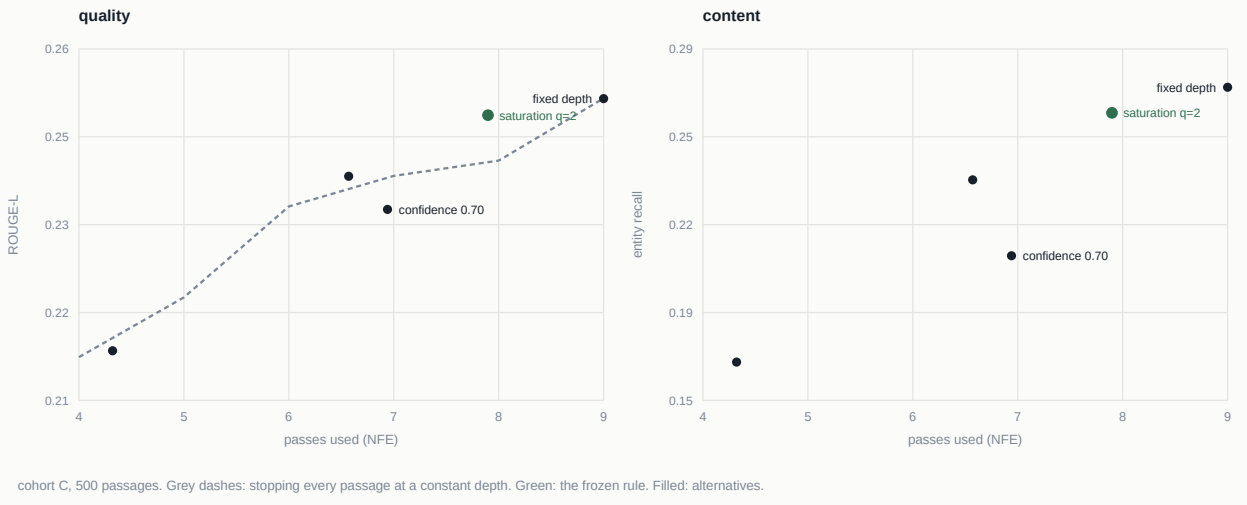


Figure 2: Every arm on the quality and content planes at once. Left: ROUGE-L against mean passes. Right: entity recall against mean passes. The rule sits above the constant-depth curve on quality, and the same arms fall further on content, because recall drops faster than ROUGE.

Table 4: The same nine passes under three meters, on the three XSum cohorts. Saved is the share of the ladder an oracle skips under that meter. The ROUGE-L meter compares a prefix with the reference at the tolerance of Section 3.2; the last column is its paired change against fixed depth on cohort C. The content meter compares a prefix with the model's own final prefix and tolerates the stated share of content words, so it is reference-free and it counts a synonym swap as lost content.

meter	A	B	C	ΔROUGE-L (C)
ROUGE-L against reference, 0.5 point	37.7%	39.3%	37.6%	+1.52
content words against final, 1%	6.4%	6.2%	7.4%	−0.08
content words against final, 10%	14.1%	13.8%	15.7%	−0.25

4.5 The last pass replaces one word in seven

The rule loses names because of what the last pass does, not because of what the rule watches.

Between the last two passes the summary keeps its length and its vocabulary size, and one content word in seven is a different word. On cohort A the text moves from 20.5 to 20.6 words and from 11.5 to 11.5 content words, while 14.3 percent of those content words are replaced. Cohort B gives 14.6 percent and cohort C gives 14.1 percent, at the same lengths. Stemming does not recover the difference, so the change is not inflection: it is a different word.

The last pass is a substitution pass, and it is the pass the rule skips on most inputs. A substitution breaks set inclusion, so a rule that waits for the name set to stop growing must stop before the last pass or not stop at all. A name that the last pass would have replaced counts as lost when the rule stops early.

That places the ceiling on a slope rather than at a point. Table 4 reads the same nine passes with three meters.

Against the reference, ROUGE-L lets an oracle skip more than a third of the ladder on every cohort. Against the model's own final prefix, a content meter that tolerates one percent of the words lets it skip 6.2 to 7.4 percent, and at ten percent of the words 13.8 to 15.7 percent. The gap is not a tolerance artefact: ten times the tolerance buys 7.4 points on cohort A, where the ROUGE

meter is four times higher throughout. The ROUGE-L oracle reads above fixed depth because it may stop at a pass that scores above the last pass, and the last pass sometimes lowers the score.

Three families now sit on one line. Documents are complete before the ladder ends, so an exact meter permits a 54.6 percent skip. Summaries are not: the last pass replaces one word in seven, so a meter that counts words permits 7 percent and a meter that counts reference agreement permits 38 percent. Grids are never complete below the top, so every meter permits zero (Section 4.6).

The ceiling of a task is set by when its content settles. What a meter may skip is set by how much of the last pass that meter discounts.

4.6 Why a grid has no ceiling

The rule works because the thing it watches becomes quiet before the ladder ends. Names reach their final set by level 4 and then stop changing. That is an observable, visible during generation, that predicts the end of the ladder.

A grid does not have this property. On 6x6, exactness first appears at level 7, for 11.5% of boards. On 9x9 hard, exactness first appears at the top level, for 84.5% of boards. Every cell keeps changing until the board is solved, so "the content is complete" is not visible before the answer exists.

The condition is testable before any gate is built. Measure the per-level exactness or content curve. If the output never becomes complete below the top level, no exit rule can help, and the ceiling in Table 1 is zero.

This also removes a competing explanation. The frozen codebook does not cause the low text quality: at the top level, 93.8% of reference summaries and 92.6% of generated summaries reconstruct exactly over their content positions. The generator, not the tokenizer, sets the text ceiling.

5 Limitations

One model family, one text task, one decode setting. No claim about scale. The text result rests on three cohorts of 500 passages from the same corpus, and the last two are now spent. A fourth cohort would not be a fair test: the rule and its bars are known, so any further cohort measures us, not the rule.

The content budget is missed. The rule is close to free on ROUGE-L and costs about one recall point on cohort C. The two statements are both true and neither replaces the other.

Decode is greedy at temperature 0, so a seed change does not move the output. The only resampling is over passages, and the intervals above are passage bootstraps.

The saving is 12 to 13%. The rate of materially worse passages is 12 to 13%. A reader who needs every passage to hold its quality should not use this rule.

The document task has a tokenizer-level ceiling and no generator run: 0.455 of held-out documents rebuild exactly at 6 of 10 codes, against 0.480 in sample, with a top level of 1.000 and an oracle of 4.535 of 10 passes. The claim for that task stops at "the codes allow it". Two thirds of the ceiling is also unclaimed: no signal here reaches better than a third of the available saving.

6 Conclusion

A next-concept model runs a fixed ladder. We measured how much of it a task lets a rule skip: 37.7 to 39.3 percent on summaries, 6.7 percent on 6x6 grids, nothing on 9x9 grids. Entity saturation

takes a third of the summarisation ceiling at 0.20 to 0.62 ROUGE-L points, and one entity-recall point on a third cohort. ROUGE-L does not report that cost.

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